

1. Identification

Product identifier 866-1520 CHROMA-CHEM® TRANSPARENT BROWN OXIDE

Other means of identification

SAP Specification 000000139037

Recommended use Non-aqueous colorant

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Company Chromaflo Technologies Corporation
2600 Michigan Avenue
Ashtabula, OH, USA 44005-0816

Canadian facility Chromaflo Technologies Canada
235 Orenda Road
Brampton, Ontario, Canada L6T-1E6

US telephone 440-997-5137

Canadian telephone 905-451-3810

NA: EMERGENCY # (3E) 866-519-4752

GLOBAL: EMERG. # (3E) (+1) 760-476-3962

3E CONTRACT # 12154

3E ACCESS CODE 334294

CANADA: CANUTEC 613-996-6666

EMERGENCY NUMBER

Product Regulatory Services ehs_americas@chromaflo.com

Supplier Not available.

2. Hazard(s) identification

Physical hazards	Flammable liquids	Category 3
Health hazards	Acute toxicity, dermal	Category 4
	Skin corrosion/irritation	Category 2
	Serious eye damage/eye irritation	Category 2A
	Sensitization, skin	Category 1
	Carcinogenicity	Category 2
	Reproductive toxicity	Category 2
	Specific target organ toxicity, repeated exposure	Category 1 (central nervous system)

Label elements



Signal word Danger

Hazard statement	Flammable liquid and vapor. Harmful in contact with skin. Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation. Suspected of causing cancer. Suspected of damaging fertility or the unborn child. Causes damage to organs (central nervous system) through prolonged or repeated exposure.
Precautionary statement	
Prevention	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Keep container tightly closed. Ground and bond container and receiving equipment. Use explosion-proof electrical/ventilating/lighting equipment. Use non-sparking tools. Take action to prevent static discharges. Do not breathe mist or vapor. Wash thoroughly after handling. Do not eat, drink or smoke when using this product. Contaminated work clothing should not be allowed out of the workplace. Wear protective gloves/protective clothing/eye protection/face protection.
Response	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. IF exposed or concerned: Get medical advice/attention. Call a POISON CENTER/doctor if you feel unwell. If skin irritation or rash occurs: Get medical advice/attention. If eye irritation persists: Get medical advice/attention. Take off contaminated clothing and wash it before reuse. In case of fire: Use appropriate media to extinguish.
Storage	Store in a well-ventilated place. Keep cool. Store locked up.
Disposal	Dispose of contents/container in accordance with local/regional/national/international regulations.
Other hazards	None known.
Supplemental information	90.918% of the mixture consists of component(s) of unknown acute dermal toxicity. If product is in liquid or paste form, hazards related to dust are not considered significant. But product may contain substances that could be potential hazards if caused to become airborne due to abrasive processes.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
Ferric Oxide		1309-37-1	20 - 40
Stoddard Solvent		8052-41-3	10 - 20
Solvent naphtha (petroleum), light aliph.; Low boiling point naphtha		64742-89-8	2.5 - 10
Isobutanol		78-83-1	1 - 2.5
Iso-butyl Acetate		110-19-0	1 - 2.5
N-butyl Acetate		123-86-4	1 - 2.5
N-butyl Alcohol		71-36-3	1 - 2.5
Solvent Naphtha (petroleum), medium aliphatic		64742-88-7	1 - 2.5
Xylene		1330-20-7	1 - 2.5
2-butanone oxime; ethyl methyl ketoxime; ethyl methyl ketone oxime		96-29-7	0.1 - 1
Ethylbenzene		100-41-4	0.1 - 1
Other components below reportable levels			20 - 40

All concentrations are in percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

4. First-aid measures

Inhalation	Move to fresh air. Call a physician if symptoms develop or persist.
Skin contact	Remove contaminated clothing immediately and wash skin with soap and water. Get medical advice/attention if you feel unwell. In case of eczema or other skin disorders: Seek medical attention and take along these instructions. Wash contaminated clothing before reuse.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	Rinse mouth. Get medical advice/attention if you feel unwell.
Most important symptoms/effects, acute and delayed	Narcosis. Behavioral changes. Decrease in motor functions. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Skin irritation. May cause redness and pain. May cause an allergic skin reaction. Dermatitis. Rash. Prolonged exposure may cause chronic effects.

Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Thermal burns: Flush with water immediately. While flushing, remove clothes which do not adhere to affected area. Call an ambulance. Continue flushing during transport to hospital. Keep victim warm. Keep victim under observation. Symptoms may be delayed.
General information	Take off all contaminated clothing immediately. IF exposed or concerned: Get medical advice/attention. If you feel unwell, seek medical advice (show the label where possible). Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves. Show this safety data sheet to the doctor in attendance. Wash contaminated clothing before reuse.
5. Fire-fighting measures	
Suitable extinguishing media	Water fog. Foam. Dry chemical powder. Carbon dioxide (CO ₂).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Vapors may form explosive mixtures with air. Vapors may travel considerable distance to a source of ignition and flash back. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Self-contained breathing apparatus and full protective clothing must be worn in case of fire.
Fire fighting equipment/instructions	In case of fire and/or explosion do not breathe fumes. Move containers from fire area if you can do so without risk.
Specific methods	Use standard firefighting procedures and consider the hazards of other involved materials.
General fire hazards	Flammable liquid and vapor.
6. Accidental release measures	
Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. Take precautionary measures against static discharge. Use only non-sparking tools. Large Spills: Stop the flow of material, if this is without risk. Dike the spilled material, where this is possible. Use a non-combustible material like vermiculite, sand or earth to soak up the product and place into a container for later disposal. Following product recovery, flush area with water. Small Spills: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Never return spills to original containers for re-use. Put material in suitable, covered, labeled containers. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid discharge into drains, water courses or onto the ground.
7. Handling and storage	
Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not handle, store or open near an open flame, sources of heat or sources of ignition. Protect material from direct sunlight. Explosion-proof general and local exhaust ventilation. Take precautionary measures against static discharges. All equipment used when handling the product must be grounded. Use non-sparking tools and explosion-proof equipment. Do not breathe mist or vapor. Avoid contact with eyes, skin, and clothing. When using, do not eat, drink or smoke. Pregnant or breastfeeding women must not handle this product. Should be handled in closed systems, if possible. Wear appropriate personal protective equipment. Wash hands thoroughly after handling. Wash contaminated clothing before reuse. Observe good industrial hygiene practices.
Conditions for safe storage, including any incompatibilities	Store locked up. Keep away from heat, sparks and open flame. Prevent electrostatic charge build-up by using common bonding and grounding techniques. Store in a cool, dry place out of direct sunlight. Store in original tightly closed container. Store in a well-ventilated place. Keep in an area equipped with sprinklers. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

US. ACGIH Threshold Limit Values

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	TWA	20 ppm	Respirable fraction.
Ferric Oxide (CAS 1309-37-1)	TWA	5 mg/m3	
Isobutanol (CAS 78-83-1)	TWA	50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	STEL	150 ppm	
N-butyl Acetate (CAS 123-86-4)	TWA	50 ppm	Non-aerosol.
	STEL	150 ppm	
N-butyl Alcohol (CAS 71-36-3)	TWA	50 ppm	
	TWA	20 ppm	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	TWA	200 mg/m3	
Stoddard Solvent (CAS 8052-41-3)	TWA	100 ppm	
Xylene (CAS 1330-20-7)	STEL	150 ppm	
	TWA	100 ppm	

Canada. Alberta OELs (Occupational Health & Safety Code, Schedule 1, Table 2)

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	STEL	543 mg/m3	Respirable.
	TWA	125 ppm	
		434 mg/m3	
Ferric Oxide (CAS 1309-37-1)	TWA	100 ppm	
		5 mg/m3	
Isobutanol (CAS 78-83-1)	TWA	152 mg/m3	
		50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	TWA	713 mg/m3	
		150 ppm	
N-butyl Acetate (CAS 123-86-4)	STEL	950 mg/m3	
		200 ppm	
	TWA	713 mg/m3	
		150 ppm	
N-butyl Alcohol (CAS 71-36-3)	TWA	60 mg/m3	
		20 ppm	
Solvent naphtha (petroleum), light aliph.; Low boiling point naphtha (CAS 64742-89-8)	TWA	1590 mg/m3	
		400 ppm	
Stoddard Solvent (CAS 8052-41-3)	TWA	572 mg/m3	
		100 ppm	
Xylene (CAS 1330-20-7)	STEL	651 mg/m3	
	TWA	150 ppm	
		434 mg/m3	
		100 ppm	

Canada. British Columbia OELs. (Occupational Exposure Limits for Chemical Substances, Occupational Health and Safety Regulation 296/97, as amended)

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	TWA	20 ppm	
Ferric Oxide (CAS 1309-37-1)	STEL	10 mg/m3	Fume.
	TWA	5 mg/m3	Dust.
		5 mg/m3	Fume.
		3 mg/m3	Respirable fraction.
		10 mg/m3	Total dust.
Isobutanol (CAS 78-83-1)	TWA	50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	TWA	150 ppm	
N-butyl Acetate (CAS 123-86-4)	TWA	20 ppm	
N-butyl Alcohol (CAS 71-36-3)	Ceiling	30 ppm	
	TWA	15 ppm	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	TWA	200 mg/m3	Non-aerosol.
Stoddard Solvent (CAS 8052-41-3)	STEL	580 mg/m3	
	TWA	290 mg/m3	
Xylene (CAS 1330-20-7)	STEL	150 ppm	
	TWA	100 ppm	

Canada. Manitoba OELs (Reg. 217/2006, The Workplace Safety And Health Act)

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	TWA	20 ppm	
Ferric Oxide (CAS 1309-37-1)	TWA	5 mg/m3	Respirable fraction.
Isobutanol (CAS 78-83-1)	TWA	50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	STEL	150 ppm	
	TWA	50 ppm	
N-butyl Acetate (CAS 123-86-4)	STEL	150 ppm	
	TWA	50 ppm	
N-butyl Alcohol (CAS 71-36-3)	TWA	20 ppm	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	TWA	200 mg/m3	Non-aerosol.
Stoddard Solvent (CAS 8052-41-3)	TWA	100 ppm	
Xylene (CAS 1330-20-7)	STEL	150 ppm	
	TWA	100 ppm	

Canada. Ontario OELs. (Control of Exposure to Biological or Chemical Agents)

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	TWA	20 ppm	
Ferric Oxide (CAS 1309-37-1)	TWA	5 mg/m3	Respirable fraction.
Isobutanol (CAS 78-83-1)	TWA	50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	STEL	187 ppm	
	TWA	150 ppm	
N-butyl Acetate (CAS 123-86-4)	STEL	200 ppm	
	TWA	150 ppm	

Canada. Ontario OELs. (Control of Exposure to Biological or Chemical Agents)

Components	Type	Value	Form
N-butyl Alcohol (CAS 71-36-3)	TWA	20 ppm	Non-aerosol.
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	TWA	200 mg/m3	
Stoddard Solvent (CAS 8052-41-3)	TWA	100 ppm	
Xylene (CAS 1330-20-7)	STEL	150 ppm	
	TWA	100 ppm	

Canada. Quebec OELs. (Ministry of Labor - Regulation Respecting the Quality of the Work Environment)

Components	Type	Value	Form
Ethylbenzene (CAS 100-41-4)	STEL	543 mg/m3	Dust and fume.
		125 ppm	
	TWA	434 mg/m3	
		100 ppm	
Ferric Oxide (CAS 1309-37-1)	TWA	5 mg/m3	Total dust.
		10 mg/m3	
Isobutanol (CAS 78-83-1)	TWA	152 mg/m3	
		50 ppm	
Iso-butyl Acetate (CAS 110-19-0)	TWA	713 mg/m3	
N-butyl Acetate (CAS 123-86-4)	STEL	150 ppm	
		950 mg/m3	
	TWA	200 ppm	
		713 mg/m3	
		150 ppm	
N-butyl Alcohol (CAS 71-36-3)	Ceiling	152 mg/m3	
		50 ppm	
Solvent naphtha (petroleum), light aliph.; Low boiling point naphtha (CAS 64742-89-8)	TWA	1590 mg/m3	
		400 ppm	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	TWA	1590 mg/m3	
		400 ppm	
Stoddard Solvent (CAS 8052-41-3)	TWA	525 mg/m3	
		100 ppm	
Xylene (CAS 1330-20-7)	STEL	651 mg/m3	
		150 ppm	
	TWA	434 mg/m3	
		100 ppm	

Biological limit values
ACGIH Biological Exposure Indices

Components	Value	Determinant	Specimen	Sampling Time
Ethylbenzene (CAS 100-41-4)	0.15 g/g	Sum of mandelic acid and phenylglyoxylic acid	Creatinine in urine	*
Xylene (CAS 1330-20-7)	1.5 g/g	Methylhippuric acids	Creatinine in urine	*

* - For sampling details, please see the source document.

Exposure guidelines Occupational Exposure Limits are not relevant to the current physical form of the product.

Canada - Alberta OELs: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

Canada - British Columbia OELs: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

Canada - Manitoba OELs: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

Canada - Ontario OELs: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

Canada - Quebec OELs: Skin designation

N-butyl Alcohol (CAS 71-36-3)

Can be absorbed through the skin.

Canada - Saskatchewan OELs: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

US ACGIH Threshold Limit Values: Skin designation

Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)

Can be absorbed through the skin.

Appropriate engineering controls

Explosion-proof general and local exhaust ventilation. Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station. Eye wash fountain and emergency showers are recommended.

Individual protection measures, such as personal protective equipment**Eye/face protection**

Wear safety glasses with side shields (or goggles).

Skin protection**Hand protection**

Wear appropriate chemical resistant gloves.

Other

Wear appropriate chemical resistant clothing. Use of an impervious apron is recommended.

Respiratory protection

If engineering controls do not maintain airborne concentrations below recommended exposure limits (where applicable) or to an acceptable level (in countries where exposure limits have not been established), an approved respirator must be worn. Use a NIOSH/MSHA approved respirator if there is a risk of exposure to vapor/mist at levels exceeding the exposure limits.

Thermal hazards

Wear appropriate thermal protective clothing, when necessary.

General hygiene considerations

Observe any medical surveillance requirements. When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Contaminated work clothing should not be allowed out of the workplace.

9. Physical and chemical properties**Appearance****Physical state**

Liquid.

Form

Liquid. Paste.

Color

Brown

Odor

Petroleum distillate odor.

Odor threshold

Not available.

pH

Not available.

Melting point/freezing point

Not available.

Initial boiling point and boiling range

Not available.

Flash point

82.00 °F (27.78 °C)

Evaporation rate

Not available.

Flammability (solid, gas)

Not applicable.

Upper/lower flammability or explosive limits**Explosive limit - lower (%)**

Not available.

Explosive limit - upper (%)	Not available.
Vapor pressure	Not available.
Vapor density	Not available.
Relative density	1.8
Solubility(ies)	
Solubility (water)	Not available.
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	Not available.
Decomposition temperature	Not available.
Viscosity	Not available.

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	Hazardous polymerization does not occur.
Conditions to avoid	Avoid heat, sparks, open flames and other ignition sources. Avoid temperatures exceeding the flash point. Contact with incompatible materials.
Incompatible materials	Strong acids. Strong oxidizing agents. Halogens.
Hazardous decomposition products	No hazardous decomposition products are known.

11. Toxicological information

Information on likely routes of exposure

Inhalation	May cause damage to organs through prolonged or repeated exposure by inhalation.
Skin contact	Harmful in contact with skin. Causes skin irritation. May cause an allergic skin reaction.
Eye contact	Causes serious eye irritation.
Ingestion	Expected to be a low ingestion hazard.

Symptoms related to the physical, chemical and toxicological characteristics	Narcosis. Behavioral changes. Decrease in motor functions. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Skin irritation. May cause redness and pain. May cause an allergic skin reaction. Dermatitis. Rash.
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Information on toxicological effects

Acute toxicity	Harmful in contact with skin.
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Components	Species	Test Results
Ethylbenzene (CAS 100-41-4)		
<u>Acute</u>		
Oral		
LD50	Rat	3500 mg/kg
Isobutanol (CAS 78-83-1)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	3392 mg/kg
Oral		
LD50	Rat	2.46 g/kg
N-butyl Alcohol (CAS 71-36-3)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	3400 mg/kg
Oral		
LD50	Rat	790 mg/kg

Components	Species	Test Results
Xylene (CAS 1330-20-7)		
<u>Acute</u>		
Oral		
LD50	Rat	3523 - 8600 mg/kg
* Estimates for product may be based on additional component data not shown.		
Skin corrosion/irritation	Causes skin irritation.	
Serious eye damage/eye irritation	Causes serious eye irritation.	
Respiratory or skin sensitization		
Canada - Alberta OELs: Irritant		
Isobutanol (CAS 78-83-1)	Irritant	
Iso-butyl Acetate (CAS 110-19-0)	Irritant	
N-butyl Acetate (CAS 123-86-4)	Irritant	
N-butyl Alcohol (CAS 71-36-3)	Irritant	
Respiratory sensitization	Not a respiratory sensitizer.	
Skin sensitization	May cause an allergic skin reaction.	
Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.	
Carcinogenicity	Suspected of causing cancer.	
ACGIH Carcinogens		
Ethylbenzene (CAS 100-41-4)	A3 Confirmed animal carcinogen with unknown relevance to humans.	
Ferric Oxide (CAS 1309-37-1)	A4 Not classifiable as a human carcinogen.	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	A3 Confirmed animal carcinogen with unknown relevance to humans.	
Xylene (CAS 1330-20-7)	A4 Not classifiable as a human carcinogen.	
Canada - Manitoba OELs: carcinogenicity		
Ethylbenzene (CAS 100-41-4)	Confirmed animal carcinogen with unknown relevance to humans.	
Ferric Oxide (CAS 1309-37-1)	Not classifiable as a human carcinogen.	
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)	Confirmed animal carcinogen with unknown relevance to humans.	
Xylene (CAS 1330-20-7)	Not classifiable as a human carcinogen.	
IARC Monographs. Overall Evaluation of Carcinogenicity		
Ethylbenzene (CAS 100-41-4)	2B Possibly carcinogenic to humans.	
Ferric Oxide (CAS 1309-37-1)	3 Not classifiable as to carcinogenicity to humans.	
Stoddard Solvent (CAS 8052-41-3)	3 Not classifiable as to carcinogenicity to humans.	
Xylene (CAS 1330-20-7)	3 Not classifiable as to carcinogenicity to humans.	
Reproductive toxicity	Components in this product have been shown to cause birth defects and reproductive disorders in laboratory animals. Suspected of damaging fertility or the unborn child.	
Specific target organ toxicity - single exposure	Not classified.	
Specific target organ toxicity - repeated exposure	Causes damage to organs (central nervous system) through prolonged or repeated exposure.	
Aspiration hazard	Not an aspiration hazard.	
Chronic effects	Causes damage to organs through prolonged or repeated exposure. Prolonged exposure may cause chronic effects.	

12. Ecological information

Ecotoxicity	The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.		
Product	Species		Test Results
866-1520 CHROMA-CHEM® TRANSPARENT BROWN OXIDE			
Aquatic			
Crustacea	EC50	Daphnia	39.1707 mg/l, 48 hours estimated
Fish	LC50	Fish	82.4551 mg/l, 96 hours estimated

Components	Species	Test Results
2-butanone oxime; ethyl methyl ketoxime; ethyl methyl ketone oxime (CAS 96-29-7)		
Aquatic		
Fish	LC50	Fathead minnow (Pimephales promelas) 777 - 914 mg/l, 96 hours
Ethylbenzene (CAS 100-41-4)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia magna) 1.37 - 4.4 mg/l, 48 hours
Fish	LC50	Fathead minnow (Pimephales promelas) 7.5 - 11 mg/l, 96 hours
Isobutanol (CAS 78-83-1)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia pulex) 950 - 1200 mg/l, 48 hours
Fish	LC50	Bleak (Alburnus alburnus) 1000 - 3000 mg/l, 96 hours
N-butyl Acetate (CAS 123-86-4)		
Aquatic		
Fish	LC50	Fathead minnow (Pimephales promelas) 17 - 19 mg/l, 96 hours
N-butyl Alcohol (CAS 71-36-3)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia magna) 1897 - 2072 mg/l, 48 hours
Fish	LC50	Bluegill (Lepomis macrochirus) 100 - 500 mg/l, 96 hours
Solvent naphtha (petroleum), light aliph.; Low boiling point naphtha (CAS 64742-89-8)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia pulex) 2.7 - 5.1 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss) 8.8 mg/l, 96 hours
		8.8 mg/l, 96 hours
Solvent Naphtha (petroleum), medium aliphatic (CAS 64742-88-7)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia pulex) 2.7 - 5.1 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss) 8.8 mg/l, 96 hours
		8.8 mg/l, 96 hours
Xylene (CAS 1330-20-7)		
Aquatic		
Fish	LC50	Bluegill (Lepomis macrochirus) 7.711 - 9.591 mg/l, 96 hours

* Estimates for product may be based on additional component data not shown.

Persistence and degradability

Bioaccumulative potential

Partition coefficient n-octanol / water (log Kow)

Ethylbenzene	3.15
Isobutanol	0.76
Iso-butyl Acetate	1.78
N-butyl Acetate	1.78
N-butyl Alcohol	0.88
Stoddard Solvent	3.16 - 7.15
Xylene	3.12 - 3.2

Mobility in soil No data available.

Other adverse effects No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Disposal instructions Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Dispose of contents/container in accordance with local/regional/national/international regulations.

Local disposal regulations Dispose in accordance with all applicable regulations.

Hazardous waste code	The waste code should be assigned in discussion between the user, the producer and the waste disposal company.
Waste from residues / unused products	Dispose of in accordance with local regulations. Empty containers or liners may retain some product residues. This material and its container must be disposed of in a safe manner (see: Disposal instructions).
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

TDG

UN number	UN1263
UN proper shipping name	PAINT RELATED MATERIAL
Transport hazard class(es)	
Class	3
Subsidiary risk	-
Packing group	III
Environmental hazards	Not available.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

IATA

UN number	UN1263
UN proper shipping name	Paint related material
Transport hazard class(es)	
Class	3
Subsidiary risk	-
Packing group	III
Environmental hazards	No.
ERG Code	3L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	UN1263
UN proper shipping name	PAINT RELATED MATERIAL
Transport hazard class(es)	
Class	3
Subsidiary risk	-
Packing group	III
Environmental hazards	
Marine pollutant	No.
EmS	F-E, <u>S</u> -E
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not established.

IATA; IMDG; TDG



15. Regulatory information

Canadian regulations	This product has been classified in accordance with the hazard criteria of the HPR and the SDS contains all the information required by the HPR.
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Controlled Drugs and Substances Act

Not regulated.

Export Control List (CEPA 1999, Schedule 3)

Not listed.

Greenhouse Gases

Not listed.

Ontario. Toxic Substances. Toxic Reduction Act, 2009. Regulation 455/09 (July 1, 2011)

Ethylbenzene (CAS 100-41-4)

Xylene (CAS 1330-20-7)

Precursor Control Regulations

Not regulated.

International regulations

Additional information is given in the Safety Data Sheet.

Stockholm Convention

Not applicable.

Rotterdam Convention

Not applicable.

Kyoto protocol

Not applicable.

Montreal Protocol

Not applicable.

Basel Convention

Not applicable.

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	No
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	No
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	No
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	No
Korea	Existing Chemicals List (ECL)	No
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	No
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes
Taiwan	Taiwan Toxic Chemicals Substances Control Act	No

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information**Issue date** 05-09-2017**Revision date** 09-10-2018**Version #** 02**Disclaimer**

The information contained herein is based on data believed to be reliable and the manufacturer disclaims any liability incurred from the use or reliance upon the same. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text. This safety information is not a license to use this material as claimed by any patents of third parties. The user alone must finally determine whether a contemplated use of this material will infringe any such patents, and for obtaining any required licenses.

Revision information

Product and Company Identification: Product and Company Identification
Hazard(s) identification: Prevention
Hazard(s) identification: Response
Hazard(s) identification: Supplemental information
Composition/information on ingredients: Composition comments
Composition/information on ingredients: Component information
Accidental release measures: Personal precautions, protective equipment and emergency procedures
Handling and storage: Precautions for safe handling
Toxicological information: Mutagenicity
HazReg Data: International Inventories
GHS: Classification